Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/18/2005
PI Name:	Nagourney, Warren Ph.D.		
Project Title:	Ultrahigh Resolution Optical Frequency Standard Using Trapped Atoms and Single Ions		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98195	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	04/01/2004	**End Date:**	01/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dehmelt, Hans (University of Washington) Fortson, Edward Norval (University of Washington)		
Grant/Contract No.:	GSN4495		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	We propose to construct an optical frequency standard based upon a single, laser-cooled, trapped indium ion. The proposed standard, whose absolute accuracy should approach one part in 10^18, would benefit from the relatively vibration-free environment of space and from the increasing compactness and efficiency of narrowband sources of radiation in the UV region of the spectrum. Recently, terrestrial sources have demonstrated sub-Hertz linewidths over reasonable times, but the microgravity environment should make this easier to achieve. Recent advances in coherently relating an optical frequency to a microwave frequency should allow one to convert the frequency standard into a time standard without loss of accuracy. This eliminates the chief objection to the use of an optical frequency as a primary time standard: the difficulty in coherently relating a frequency of about 1e15 Hz to a microwave frequency which can be electronically counted and which therefore can be used to define the second. The advantages of optical standards with their much higher intrinsic Q and potentially smaller systematic errors (when compared to microwave standards) gives an optical standard a new lease on life as a promising candidate for the future primary time standard of choice. The task for the first year of the project will be to improve the linewidth of the recently-observed resonances of the forbidden transition to several 10's of Hz and the second year's task will be the construction of a compact, solid-state frequency-quadrupled cooling laser, a "twin" of the existing compact "clock" laser currently being used for spectroscopy. With two solid-state laser systems, the entire optical frequency standard could be quite compact and would consume a fairly small amount of electrical power; these qualities would be very favorable for the potential use of the standard in space. An extremely stable optical frequency standard has many possible applications in the areas of metrology, tests of general relativity and the equivalence principle, detection of gravitational waves as well as the more mundane areas of highly precise navigation and surveying. A frequency standard based upon a single indium ion could be used in a proposed experiment to measure the time rate of change of the fine structure constant. This has become particularly interesting due to the recent astronomical observation of a statistically significant change in alpha over a large fraction of the size of the universe. The proposed experiment would compare a single ion indium standard (the reference) to a single ion barium standard, whose change of frequency with alpha is about 10 times greater than in indium. The sensitivity in the fractional change of alpha is anticipated to be 5x10^-19/year.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved time metrology will have many benefits for navigation, surveying and communication.
Task Progress:	The principal task this year is to obtain a very narrow optical resonance line from a forbidden transition in a single indium ion. This line was observed several years ago using a much more primitive apparatus than we are currently using. Although the frequency was very accurately known (published error of about a kilohertz), the frequency measuring equipment formerly used had an error of about 50 MHz, requiring a search over this range. The diode laser system was based upon an external cavity diode laser which was very difficult to stabilize, since the commercial diode controller did not allow adequate electronic bandwidth to permit a frequency servo to work properly. Finally, a small and very jittery temperature-stabilized and ultra-high finesse (finesse 30000) cavity was used - the jitter was nearly 1000 times the intrinsic width of the cavity. The observation of the line was therefore both difficult to repeat and did not yield a very sharp resonance. All of these aspects of the experiment have been corrected. Frequency measurement is now based upon a series of saturated Tellurium lines, which permit reproducible setting of the laser frequency to about 100 kHz or less (more than 500 times better than the wave-meter). The diode laser has been replaced with a solid-state YAG non-planar-ring-oscillator, whose free-running line-width is 2 kHz, about 100 times better than the diode system. Finally, the laser is reliably locked to a very stable cavity whose (measured) line-width is 16 kHz and whose (measured) drift is less than 5 Hz per second in the infrared. The very capable graduate student currently working on the project has set up a number of alarms to facilitate reliable searches for the transition using our complicated apparatus, which has 6 frequency locking systems for lasers and doubling cavities. Although the tellurium lines allow reproducible resetting of the laser frequency, the initial search is over the same 50 MHz as previously (once the forbidden line is found and the tellurium spectrum is calibrated, subsequent experiments will be much easier). The search is currently under way and should shortly be successful. A fortuitous development has considerably improved the prospect for using a completely solid-state cooling laser in the near future (this would make the entire experiment solid-state, with considerable advantages for long-term reliability and stability). We had purchased a 922 nm external cavity diode laser with the hope of injection-locking a multimode high-power diode and using the combination as a Master-Oscillator-Power-Amplifier (MOPA) for frequency quadrupling into the UV (as was very successfully demonstrated with our former clock laser, which provided 10 mW in the UV). A number of attempts to injection lock a commercial diode had a mixed outcome: about 50 mW was obtained with 10 mW input, but the system was very sensitive to temperature and difficult to align. We recently purchased a commercial laser amplifier which had none of these problems and provided about 500 mW of power at 922 nm (this exceeds the specifications of the amplifier by a factor of more than 2). We are now assembling a duplicate of our frequency-quadrupled clock laser and hope to have it completed within a couple of months.
Bibliography Type:	Description: (Last Updated: 03/30/2017)
Articles in Peer-reviewed Journals	Hong, T.; Cramer, C.; Nagourney, W.; Fortson, E. N. "Optical Clocks Based Upon Ultranarrow three-Photon Resonances in Alkaline Earth Atoms" PRL volume 94, 050801 (2005) , Feb-2005